PROJECT PROPOSAL

1. Executive Summary

The Banking Transaction System is a secure, reliable, and web-based application designed to automate core banking operations. It empowers customers to manage multiple account types (Savings, Checking, and Fixed Deposit) while providing administrators with control tools to oversee accounts and monitor audit logs. The primary objective is to enhance operational efficiency, reduce manual errors, and .provide a seamless financial management experience

2. Problem Statement & Background

Traditional and manual banking administrative processes often suffer from high operational overhead, slow transaction processing, and potential human error in calculating interest rates or maintaining transaction consistency. Customers require a fast, transparent platform to execute transfers and track financial records, while banking administrators need centralized visibility over account statuses and .compliance audits without compromising system security

3. Objectives & Proposed Solution

Objectives

- **Operational Efficiency:** Streamline core operations and eliminate manual calculation errors in transaction processing and interest generation through automated system rules.
- **Customer Empowerment:** Provide a secure, 24/7 digital web platform for self-service account management, instant fund transfers, and exportable financial statements (CSV).
- **Administrative Control:** Empower bank administrators with a centralized control panel and audit logs to monitor system activity, manage account security statuses, and ensure compliance.

Proposed Solution

A full-stack web-based banking platform built with a robust **Java Spring Boot** backend .and an **SQLite** database, featuring a responsive **Web UI** for seamless user interaction

- **From a Business & Operational Perspective:** The system bridges the gap between customers and administrators by delivering an intuitive portal for instant account management and transfers, alongside a secure administrative .dashboard for tracking audit logs and account controls
- **From a Technical & Quality Perspective:** The platform applies core **Object-Oriented Programming (OOP)** principles to handle complex account hierarchies (Savings, Checking, Fixed Deposits), and guarantees system stability and reliability through automated testing frameworks (**JUnit 5**, **MockMvc**) and structured manual quality assurance strategies (**EP**, **BVA**, .(**Decision Tables**

4. Project Scope

In-Scope

- **User & Role Management:** Visitor registration, Customer authentication, and Admin access control.
- **Account Types:**
  - **Savings Account:** Minimum balance of 1,000 EGP with monthly interest calculation.
  - **Checking Account:** Overdraft facility supported down to -1,000 EGP.
  - **Fixed Deposit:** Time-locked deposits starting at 5,000 EGP with term-end interest payout.
- **Core Transactions:** Deposits, withdrawals, transfers, and real-time balance updates.
- **Reporting & Auditing:** Audit logging for admin actions and CSV statement downloads for customers.
- **User Interface & API:** Web UI for end-users and RESTful APIs for integration.

Out-of-Scope

- ATM physical interface integration and credit/debit card processing.
- Multi-currency support (System strictly operates in EGP).
- SMS and Email notification gateways.

- Mobile app native client development.

5. Methodology & Implementation Plan

The project follows an Agile development approach utilizing Jira for task management, .sprint tracking, and bug logging

1. **Requirements & Architecture Design:** Defining functional/non-functional requirements, database schemas, and API endpoints.
2. **Core Development:** Building backend services using Java Spring Boot and UI components.
3. **Quality Assurance & Testing:**
    - **Automated Unit Testing:** JUnit 5 and MockMvc for business logic verification.
    - **API Testing:** Postman collections for endpoint validation.
    - **Manual & UI Testing:** Execution of boundary and decision-table test cases.
4. **Final System Delivery:** End-to-end regression testing and final documentation.